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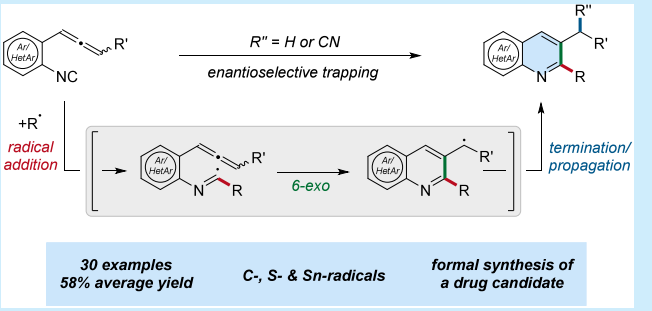
Radical Addition to 2-Allenylaryl Isocyanides for the Preparation of 2,3-Difunctionalized Quinolines

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ABSTRACT: 2-Allenylaryl isocyanides are presented as a novel platform to construct quinolines carrying various substituents in positions 2 and 3 via radical addition cyclization cascades. Carbon-, sulfur-, and tin-centered radicals add to the isonitrile functionality to generate the corresponding imidoyl radicals, which undergo 6-*exo*-cyclization onto the *ortho*-allenyl moiety. The newly generated benzylic radicals are either reduced or cyanated. In the latter case, enantioselective trapping is achieved.



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Aryl isocyanides are highly potent radical acceptors leading to imidoyl radicals upon addition.[1−4] This enables isocyanides to form—unlike other radical acceptors such as alkenes, alkynes, nitriles, etc.—1,1-adducts on the terminal carbon, thereby acting as C1-synthons.[2] The efficiency of such processes renders the isonitrile functionality a highly versatile radical acceptor to realize radical addition cyclization cascades, as initial radical addition proceeds almost exclusively on the isocyanide functionality in the presence of various other typical radical acceptors.[2] Despite being known for their great performance in radical reactions since the 1960s,[5] it was not until the 1990s that Curran introduced this functional group in radical cascade reactions,[6] laying the foundation for related strategies for the synthesis of various N-heterocyclic natural products.[7] Over the last three decades, aryl isocyanides have been intensively used in radical chemistry, enabling access to valuable products.[3,4,8,9]

In particular, *ortho*-acceptor-substituted aryl isocyanides **1** stand out by their remarkable efficiency for the construction of diverse N-heteroarenes.[9] The imidoyl radicals **2** formed after initial radical addition to the isocyanide **1** are prone to cyclize onto pending *ortho*-radical acceptors. Mainly, the nature of the *ortho*-acceptor moiety directs the reaction outcome, enabling access to a large series of important N-heterocyclic products (Scheme 1A). Those include phenanthridines (acceptor A = arene),[10] indoles (A = alkene,[11,12] alkyne[13−15]), benzimidazoles (A = N_3[16]), benzoxazoles (A = ether[17]), benzothiazoles (A = thioether[18,19]), benzoselenazoles (A = selenide[18]), quinoxalines (A = isocyanide,[20] azole[21]), quinazolines (A = CH_2N_3[22]), pyrrolopyridines (A = nitrile[23]), and dibenzodiazepines (A = aniline[24]). However, to access quinolines via analogous strategies, specially designed alkenes[25] or alkynes[26] are mandatory as acceptor functionalities to ensure efficient six-membered ring formation.[14,15]

Quinolines represent an important compound class, as various members are associated with useful biological activities, which renders the synthesis of these heterocycles a desired target (Scheme 1B). While most classical strategies exploit inter- or intramolecular cyclization of nitrogen-containing fragments with fitting carbon-based counterparts (e.g., classic Skraup or Doebner−Miller synthesis), extensive effort has been recently spent to expand the scope of accessible quinolines via mild catalytic methods.[27] Along these lines, (catalytic) radical cascade cyclizations using aryl isocyanides are generally advantageous for installing valuable functional groups in sophisticated patterns within the synthesis.[25,26]

Allenes are also efficient acceptors, with radicals adding predominantly at the central sp-carbon.[28] Thus, we envisioned that the formerly not examined *ortho*-allenylaryl isocyanides **3** might be eligible substrates for efficient quinoline synthesis, enabling simultaneous C2- and C3-benzylic functionalization (Scheme 1C). While this substrate class **3** was recently studied in Pd-catalyzed living cyclocopolymerizations,[29] to the best of our knowledge no radical transformations have been reported with these interesting acceptors so far. Initial radical addition onto isocyanide **3** would lead to imidoyl radical **4** that should engage in a 6-*exo*-cyclization to generate benzylic radical **5**, which can further react with various trapping reagents. In this

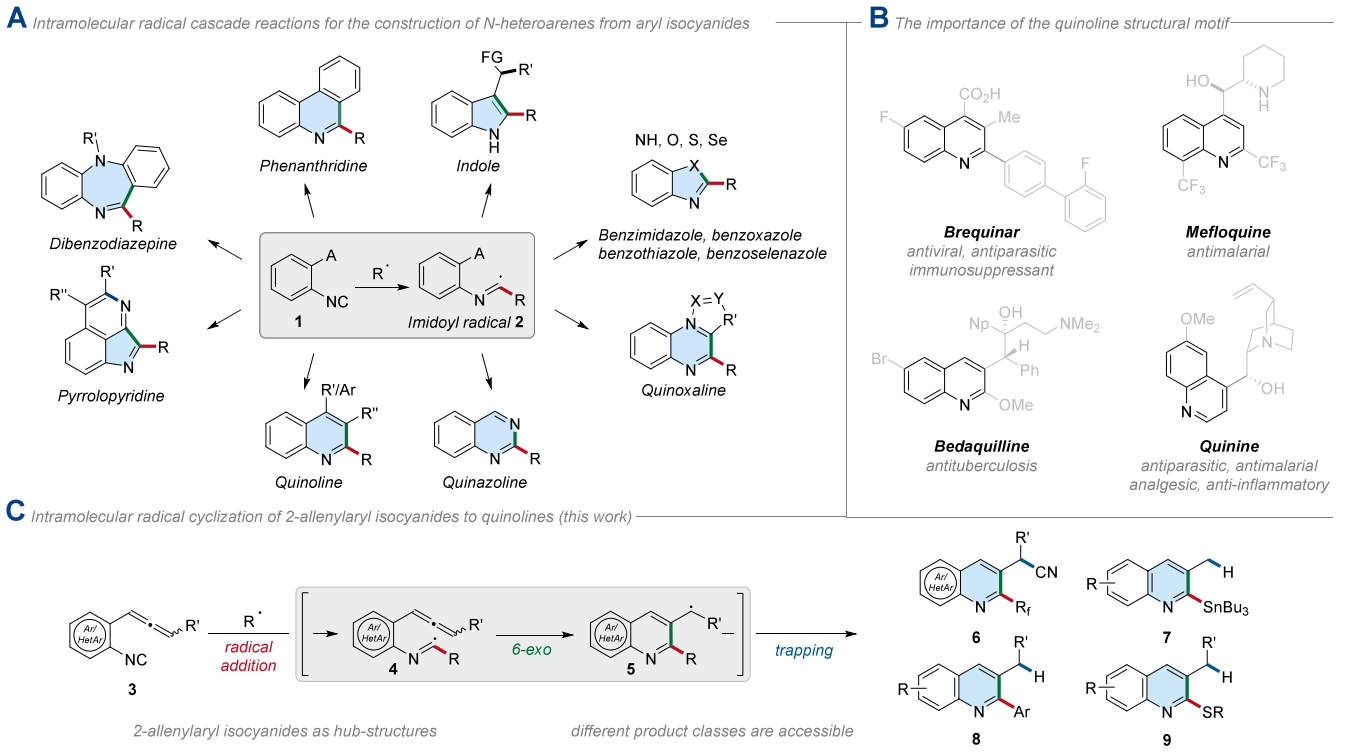
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Scheme 1. Radical Cascade Reactions with Aryl Isocyanides as Acceptors and Quinolines as Core Structures in Biologically Active Compounds[a]



[a](A) General overview of radical N-heteroarene syntheses using 2-acceptor-substituted aryl isocyanides **1**. (B) The importance of the quinoline structural motif. (C) Our approach of using 2-allenylaryl isocyanides **3** for the synthesis of quinolines **6**–**9**.

Letter, we demonstrate the use of isonitriles **3** as valuable radical acceptors for the preparation of a variety of quinolines **6**–**9**.

We initiated our studies by examining the 2-allenylaryl isocyanide **3a** (R′ = H) in a radical copper-catalyzed cyanotrifluoromethylation sequence (Scheme 2A).[30] With the Togni II reagent as a CF_3-radical precursor, $Cu(CH_3CN)_4PF_6$ as a precatalyst (5.0 mol %), **L1** as a ligand (8.5 mol %), and TMSCN (2.4 equiv) as a cyanide source in ethyl acetate, the targeted 3-quinolineacetonitrile **6a** was obtained in 73% isolated yield (Scheme 2B, entry 1). Alternative $^{\bullet}CF_3$ precursors like the Togni I or Umemoto reagents provided **6a** in significantly lower yields (Scheme 2B, entries 2 and 3). Performing the Cu-catalyzed cascade reaction without a ligand or with achiral 1,10-phenanthroline led to poor product formation and incomplete conversion of **3a** (Scheme 2B, entries 4 and 5). Acetonitrile, a commonly used solvent in related transformations,[31] was found to have a deteriorating effect on the reaction outcome (Scheme 2B, entry 6).

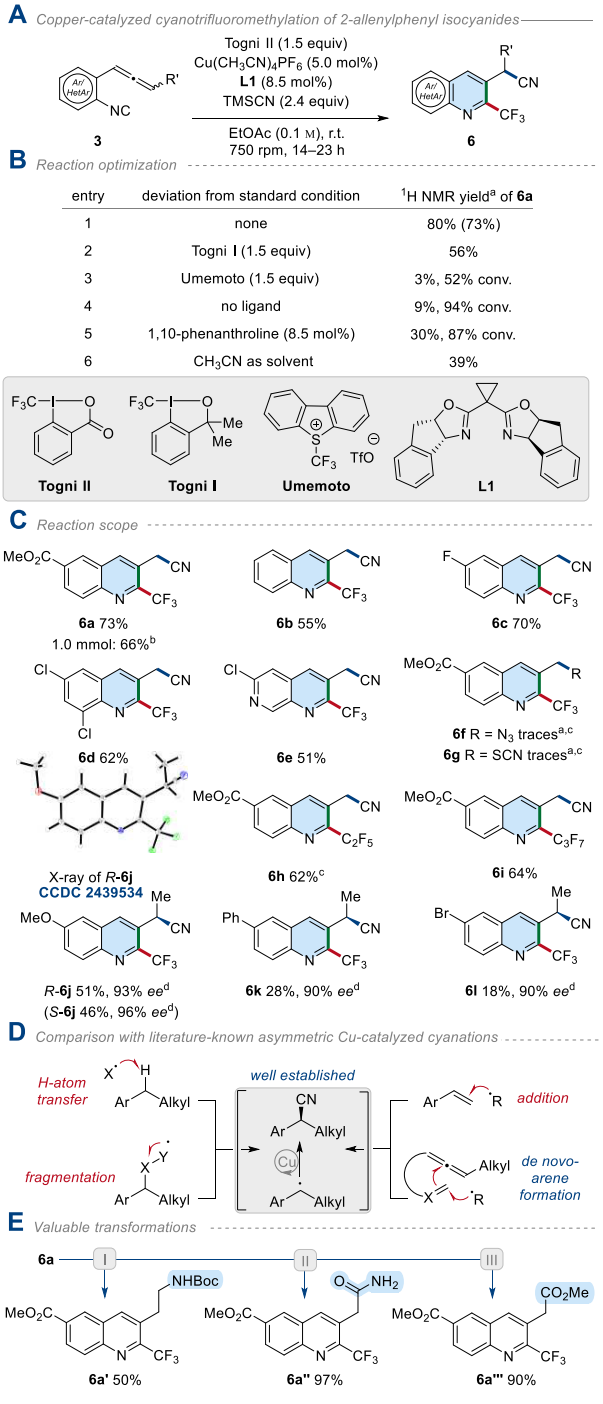
Applying the best conditions, we next examined the scope of this transformation (Scheme 2C). The unsubstituted **6b** as well as the halide derivatives **6c** and **6d** were formed in good yields (55–70%). Notably, using the pyridine-derived isocyanide **3e** allowed expansion of the product scope from simple quinolines to naphthyridineacetonitrile **6e** (51%). However, for azidation[32,33] or thiocyanation,[32,34] applying other nucleophile sources like $TMSN_3$ or TMSNCS provided only traces of the corresponding products **6f** and **6g**. Togni II derivatives carrying longer perfluoroalkyl chains were eligible radical precursors for the successful preparation of quinolines **6h** (62%) and **6i** (64%).

Allene acceptors with an additional substituent at the terminal position undergo the addition–cyclization sequence, leading to prochiral secondary benzylic radicals of type **5**, eventually affording chiral 3-quinolineacetonitriles **6** through Cu-mediated cyanation. We were very pleased to find that with the methyl-substituted 2-allenylaryl isocyanides **3j**–**l** under our standard conditions with chiral ligand **L1** the desired nitriles **6j**–**l** were formed with high enantioselectivity (90–96% ee), albeit in diminished yields (18–51%). This is likely due to competing side reactions as the more nucleophilic disubstituted allenes **3j**–**l** rapidly decompose in solution. Since more electron-rich isocyanides decompose slower,[35] the highest yield in this series was noted for the methoxy derivative **6j** (51%). Unfortunately, no further improvement could be achieved in a screening of other chiral bisoxazoline ligands (see SI section 3, Scheme S1 for details). The absolute configuration of **6j** was unambiguously assigned by X-ray crystal structure analysis (X-ray structure of the *R*-enantiomer prepared with **L1** is depicted in Scheme 2C). The absolute configurations of **6k** and **6l** were assigned in analogy. Of note, the asymmetric copper-catalyzed cyanation of secondary benzylic radicals has been heavily investigated in recent years.[31,36,37]

In this context, prochiral benzylic radicals for enantioselective cyanation were generated through benzylic hydrogen atom abstraction, fragmentation, or radical addition to styrene derivatives.[37] Thus, our current study discloses a novel synthetic route for the generation of secondary benzylic radicals through concomitant arene construction (Scheme 2D) (for a proposed mechanism, see SI chapter 4, Scheme S2). The synthetic value of the 3-quinolineacetonitrile products was demonstrated by the successful transformation of **6a** to the

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Scheme 2. Copper-Catalyzed Cyanotrifluoromethylation of 2-Allenylaryl Isocyanides 3*

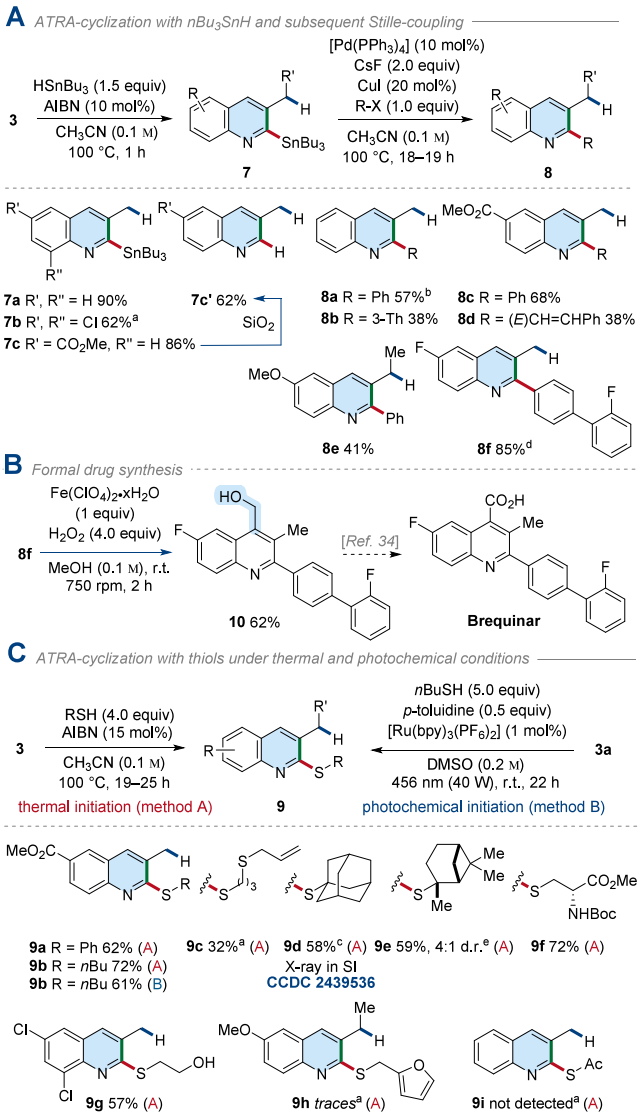


*Reactions (0.1 mmol) were conducted under an argon atmosphere, and isolated yields are reported. [a]NMR yield with CH_2Br_2 as the internal standard. Isolated yield given in parentheses. Reaction was performed on a [b]1.0 or [c]0.05 mmol scale. [d]Enantiomeric excess was determined by HPLC on chiral stationary phases. I) **6a** (0.1 mmol, 1.0 equiv), $NiCl_2{\cdot}6H_2O$ (1.5 equiv), $(Boc)_2O$ (2.0 equiv), $NaBH_4$ (7.0 equiv), MeOH (0.05 M), 0 °C → r.t., 2.5 h; II) **6a** (0.1 mmol, 1.0 equiv), (*E*/*Z*)-acetaldoxime (4.0 equiv), $Pd(OAc)_2$ (20 mol %), PPh_3 (40 mol %), EtOH/H_2O (0.07 M, 5:1), 85 °C, 6 h; III) **6a** (0.1 mmol, 1.0 equiv), H_2SO_4 (*w* = 98%, 0.2 M), MeOH (0.05 M), 100 °C, 21 h. X-ray of *R*-**6j**: Thermal ellipsoids are shown at 30% probability.

Boc-protected amine **6a′** (50%), the amide **6a″** (97%), and the ester **6a‴** (90%, Scheme 2E).

To showcase that the 2-allenylaryl isocyanides **3** serve as platforms to diversely decorated quinolines, we further examined other typical radical processes. Conditions adapted from the original Fukuyama indole synthesis[12] enabled the preparation of 2-stannylated quinolines **7a**–**c** through hydrostannylation in very good yields (62–90%) (Scheme 3A). The protodestannylated congener **7c′** was obtained in 62% yield from **7c** after flash column chromatography on SiO_2. Moreover, the 2-arylated quinoline **8c** could be prepared in 26% yield when 5.0 equiv of phenyl iodide was added, showing that iodine abstraction by the tin radical is faster than tin radical addition to the isocyanide (see SI for details). However,

Scheme 3. Scope of Reactions with 2-Allenylaryl Isocyanides 3*



*Reactions (0.1 mmol) were conducted under an argon atmosphere, and isolated yields are reported. [a1]H NMR yield with CH_2Br_2 as the internal standard. Reaction pursued on a [b]0.5 mmol or [c]0.2 mmol scale. [d]DMF (0.01 M) was added as a cosolvent in the Stille coupling. [e]Diastereomeric ratio determined by ^{1}H NMR from the crude mixture.

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the same 2-functionalized quinolines can be accessed in higher yields through one-pot Stille cross-coupling from the intermediate stannanes with various aryl iodides or vinyl iodides as demonstrated by the synthesis of quinolines **8a**−**e** (38−68%).[12] The valuable Stille cross coupling was also used successfully for the preparation of quinoline **8f** (85%) from isocyanide **3c** using the corresponding aryl triflate. With the quinoline **8f** in hand we realized the formal synthesis of brequinar by installing the C-skeleton of alcohol **10** *via* a Fenton reaction in methanol with 62% yield (Scheme 3B).[38]

Atom transfer radical addition (ATRA) cyclization with thiols[14,15] was also examined using our 2-allenylaryl isocyanides **3** as radical acceptors (Scheme 3C). Cyclizing hydrothiolation applying a thermal initiation protocol (method A) with AIBN (α,α'-azoisobutyronitrile) showed a good functional group tolerance as demonstrated by the reaction of **3a** with aromatic (**9a**) and aliphatic (**9b**−**f**) thiols. Ester and amide functionalities at the thiol component were tolerated (**9f**). However, allyl thiol provided **9c** in a moderate yield as a result of a thiol−ene reaction on the initially targeted product (see SI for details). Sterically more demanding tertiary thiols afforded the corresponding products **9d** and **9e** in comparable yields. Noteworthy, natural product-derived scaffolds like monoterpene **9e** or protected cysteine **9f** could be used as the thiol component in these transformations (59−72%). The terminal substituted allene **3j** yielded **9h** only in traces due to the lower stability of the starting material. Unfortunately, thioacetic acid was not an eligible thiyl radical precursor in this cascade reaction, and **9i** was not detected. Along with the thermal protocol, a photochemical initiation process (method B) was successfully applied for the preparation of **9b** from allene **3a** and butyl thiol (see SI for details).[39] The quinoline **9b** was obtained in slightly diminished yields (61%) compared with the thermally initiated process.

In summary, we introduced 2-allenylaryl isocyanides **3** as versatile radical acceptors for the preparation of diversely substituted quinolines **6**−**9**. 2-Trifluoromethylated 3-quinolineacetonitriles **6** were obtained through copper-catalyzed perfluoroalkylcyanation of **3**. With terminal substituted allenes, chiral 3-quinolinacetonitriles **6j**−**l** were accessed with high enantioselectivity (90−96% ee) using a chiral Cu catalyst. Furthermore, ATRA cyclization applying nBu$_3$SnH, aromatic, and aliphatic thiols provided various S- and Sn-substituted quinolines. The latter engaged in Stille cross couplings, enabling the formal synthesis of the drug brequinar. While our study examined 2-allenylaryl isocyanides **3** across four different radical protocols, each yielding distinct product classes, we anticipate that additional reaction modes could be applicable when isonitriles of type **3** are used as the starting material. Therefore, we encourage others to consider this platform in their synthetic planning.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.5c01436.

Details, characterization data, nuclear magnetic resonance (NMR) spectra, HPLC traces, and X-ray data (PDF)

Accession Codes

Deposition Numbers 2439534−2439536 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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Author Contributions

The manuscript was written through the contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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